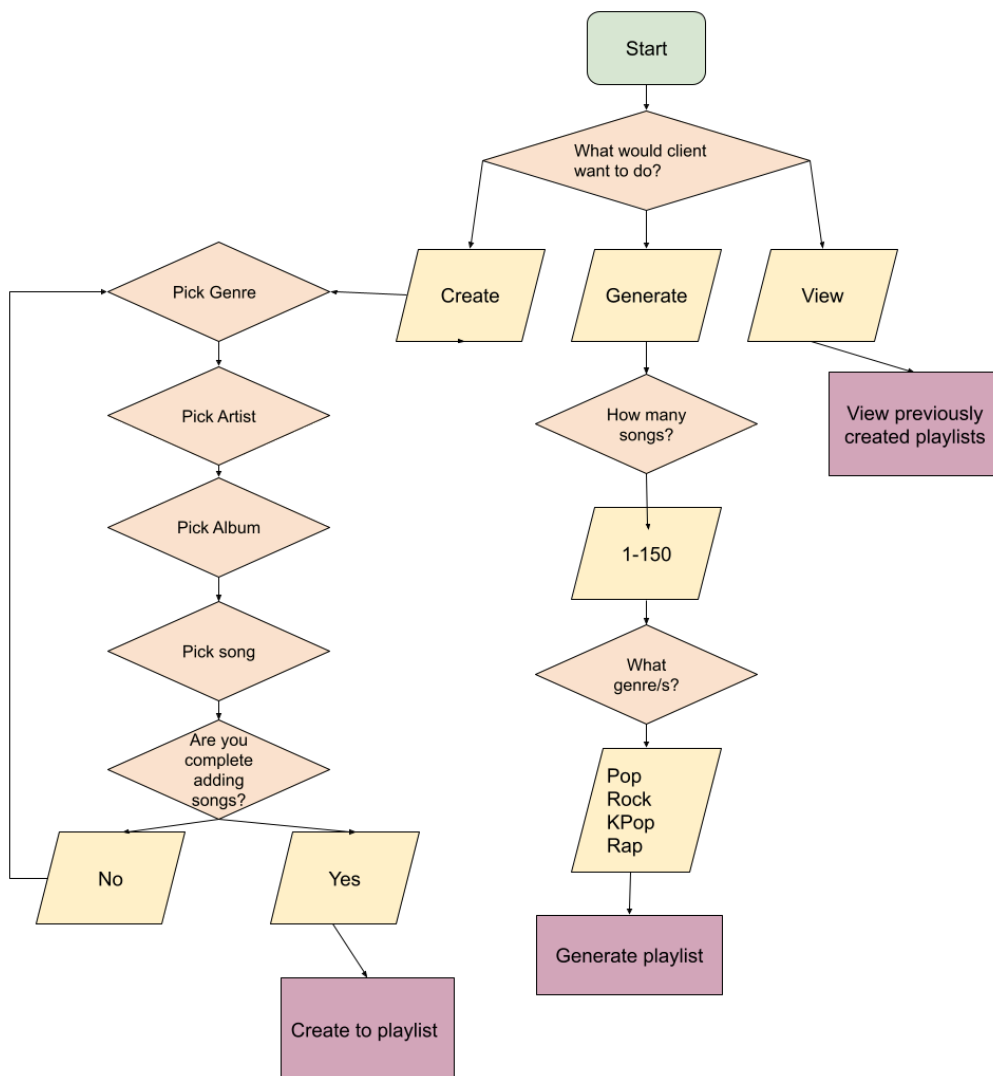


Criterion B: Design Overview & Test Plans

Design Overview

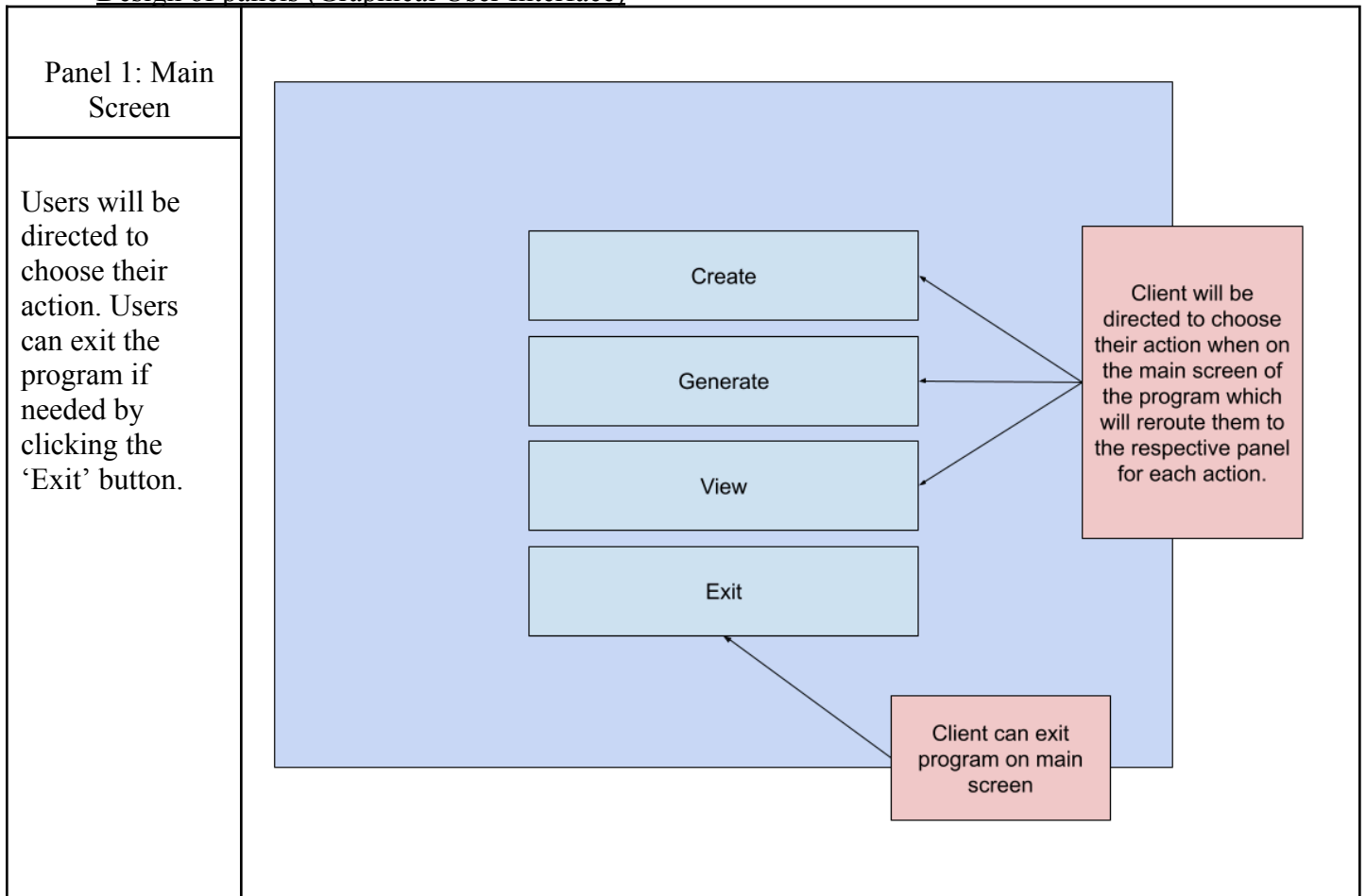
Program Overview

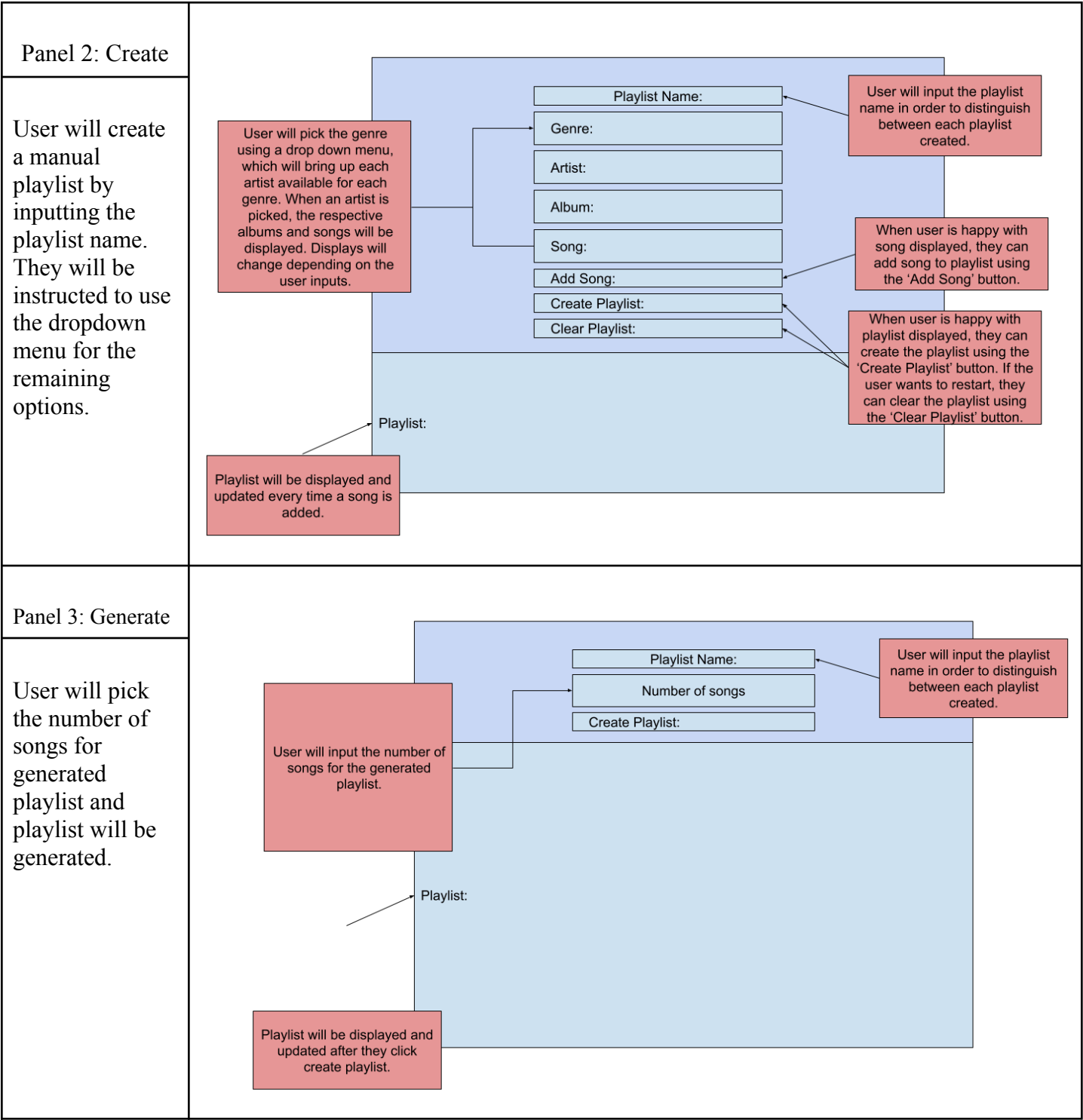


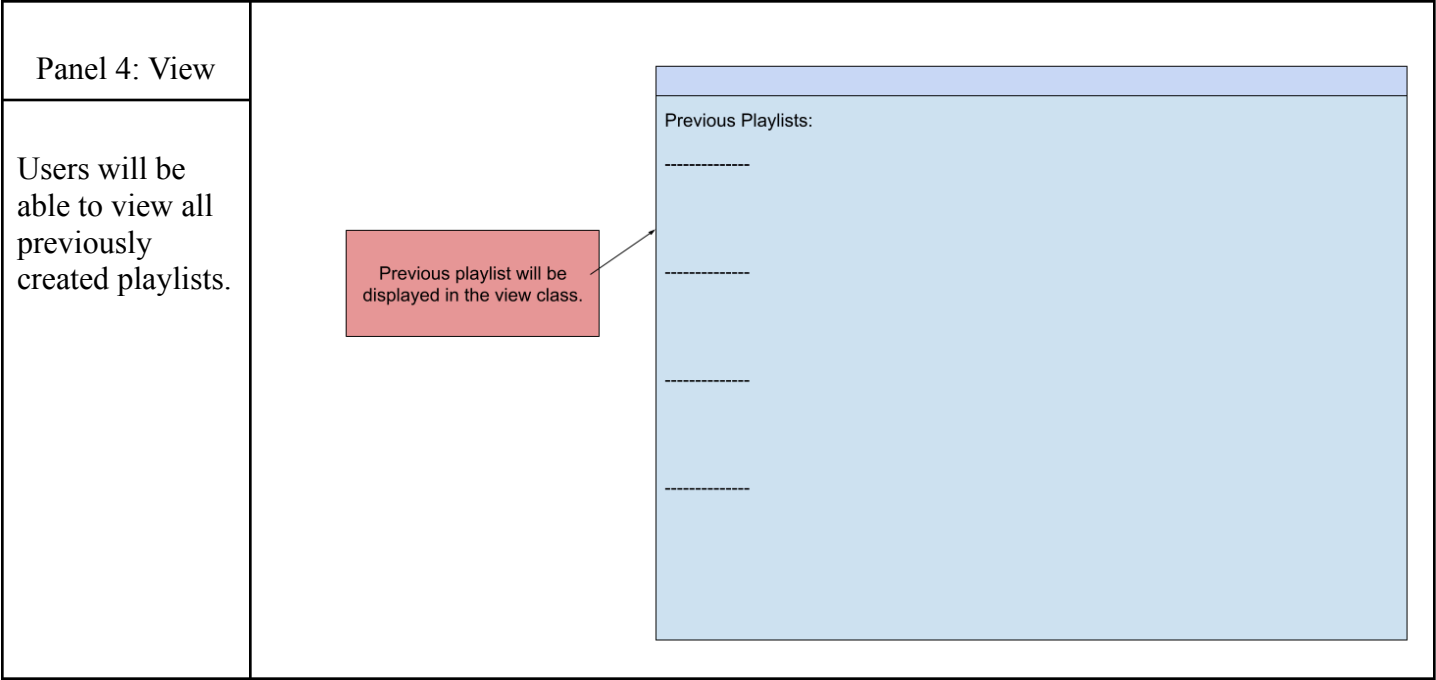
Functionality of each class:

Class	Methods	Function
Main	main(String[] args)	Base of program, the starting point that will have a structure to reroute to methods depending on user input.
Generate	generatePlaylist()	Will generate a random playlist depending on user-defined length.
	getAllAvaliableSongs()	Will read through the available songs in the music library and turn them into a list.
	savePlaylistsToFile(List)	Save the playlist to a list.
Create	createPlaylist()	Will direct the user to create their own playlist depending on the user-input.
	clearPlaylist()	Will clear the previously created playlist.
	removeSongFromPlaylist()	Removes added song from playlist.
	queueSong()	Add songs to a queue for the user to view.
	showQueue()	Queue will be outputted for the user to view.
View	loadPlaylists()	Playlists will load from a text file and then store them in a 2D array.
	displayPlaylists()	Displays the previously saved playlists in a user-friendly form

Design of panels (Graphical User Interface)







Test Plan

Method to be tested	Test Method
Test if users will be able to generate playlists with only the playlist length and genre imputed.	Attempt to create a playlist between 1-150 songs and confirm that the playlist is generated. Repeat 10 times to ensure functionality.
Test if users will be able to select genre, artist, albums, and songs, to add to the playlists.	Create playlists in order to ensure all songs are able to be added to playlists. Attempt to select genre, artist, albums, and songs, to add to the playlist, composed of various genres and artists. If the program includes multiple genres in order to provide a variety of options for the user the playlist will be able to include the various genres and artists.
Test if the program will be able to create playlists based on user input.	Ensure that the user can create various playlists. To test this, select various genres, artists, albums, and songs, in order to ensure the outcome is accurate. Will also attempt to control the playlist length of both created and generated playlists. Will add or remove songs in order to test program functionality and adaptability.
Test if the user will be able to delete created playlists.	Will create various playlists and then attempt to clear the playlist. Will also view the saved playlists before and after removing them in order to ensure the functionality of the view class.
Test if users will be able to expand playlists by adding new songs or genres.	Will create a playlist and then attempt to remove a specific song. Repeat 10 times to ensure functionality. Will view playlists before and after attempting to add or delete in order to ensure the playlist has been modified.
Test if the program will display created and saved playlists.	Create a playlist and attempt to view it. If it is shown, it has been successful.
Test if Program will be able to save a playlist created to view at a later date.	Create a playlist and attempt to save it. If it is shown in the view section, it has been successful.
Test if program will validate the user input in order to prevent user error.	Attempt to crash the program by not inputting the name of playlist, if the validation works, it will attempt to have user type in name for playlist. Also will attempt to generate a playlist without defining a length, if it does not allow the user to do so, validation works.
Test if the program will have a graphical user interface (GUI) which will allow for a better user experience.	Ensure all buttons work and do what they are intended to do. Do multiple run throughs in order to ensure functionality. In order to have a proper test, the client will do a black-box test with no prior knowledge of how to work the program in order to test if the user experience is straight forward, simple, and functional.